THE COTANGENT COMPLEX COMMUTES WITH FILTERED COLIMITS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cotangent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's 'this is true because' argument naming the two commutativity facts; both are stated in the span and the conclusion is drawn immediately.

Editorial note. The single reference of the counted spans is recorded as a non-load-bearing navigation pointer; the argument that consumes it stays inside the counted proof span.

Editorial note. Preamble: the retained preamble of this chapter (sources/stacks-cotangent/retained-preamble.tex, lines 1-63), containing the Xy-pic setup, the comment environments, the theorem declarations and the macros including \NL.

Allowed standard prerequisites (editorial).

- The forgetful functor from algebras to sets and its adjoint commute with filtered colimits. used at proof lines 380-382. Provides the compatibility for the underlying category.
- The functor of differentials commutes with filtered colimits. Algebra, Lemma algebra-lemma-colimit-differentials, used at proof lines 383-384. Provides the compatibility for the modules of differentials.
- The construction of the cotangent complex from a resolution and the differentials. used at proof lines 379-385. Is what the two compatibilities are applied to.

Source reference. Original lemma 031G: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3983–3983. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let $A_i \to B_i$ be a system of ring maps over a directed index set I. Then* $\operatorname{colim} L_{B_i/A_i} = L_{\operatorname{colim} B_i / \operatorname{colim} A_i}$.

Proof. This is true because the forgetful functor $V : A\text{-}Alg \to Sets$ and its adjoint $U : Sets \to A\text{-}Alg$ commute with filtered colimits. Moreover, the functor $B/A \mapsto \Omega_{B/A}$ does as well (Algebra, Lemma 031G). □

Date: stacks-project:cotangent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.